

MOURGASH, BulgariaWMO: **156000**Lat: **42.833N**Lon: **23.667E**Elev: **1692**StdP: **82.58**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-14.2	-20.7	0.7	-13.0	-18.0	0.9	-11.6	34.3	-1.1	33.8	-0.4	13.1	0	0.956

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	22.1	14.7	20.5	14.2	19.1	13.8	16.4	19.5	15.6	18.7	14.7	17.7	5.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	13.5	17.4	14.5	12.7	16.7	13.6	11.9	16.0	52.9	19.6	50.0	18.8	47.3	17.9	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	24.3	20.1	-18.8	25.5	2.7	2.8	-20.8	27.6	-22.3	29.2	-23.9	30.8	-25.9	32.8		
(5)			-19.1	18.6	2.7	1.2	-21.0	19.5	-22.6	20.2	-24.1	20.8	-26.0	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-4.6	-4.1	-1.4	3.3	8.1	11.6	13.8	14.1	9.9	5.6	1.8	-3.0
	DBStd	8.00	5.26	5.23	4.86	4.39	3.74	3.67	3.29	3.30	4.12	4.87	5.06	5.09
	HDD10.0	2370	451	396	353	205	81	25	6	4	50	148	248	403
	HDD18.3	5004	709	629	612	451	316	204	146	135	252	394	496	661
	CDD10.0	419	0	0	0	4	23	72	123	133	48	13	2	0
	CDD18.3	11	0	0	0	0	0	1	4	5	1	0	0	0
	CDH23.3	19	0	0	0	0	1	1	11	6	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	8.5	10.1	10.4	9.9	8.8	7.6	6.8	6.8	6.6	7.4	8.1	10.0	10.3
Precipitation	PrecAvg	737	41	37	46	55	99	92	78	61	63	55	36	41
	PrecMax	1209	93	64	102	117	205	188	199	283	207	117	84	92
	PrecMin	518	11	12	12	10	31	35	16	8	9	4	4	8
	PrecStd	168	20	17	20	29	45	42	49	56	48	31	17	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	9.2	11.5	16.4	19.0	22.3	25.1	24.3	21.7	17.6	14.9	9.6
		MCWB	2.4	4.0	5.4	9.2	13.1	15.7	14.1	15.2	13.4	10.6	7.7	2.9
	2%	DB	5.2	6.2	8.9	13.6	16.4	20.0	22.2	22.0	19.1	15.1	11.5	6.9
		MCWB	1.2	1.9	4.7	7.8	11.6	14.9	14.2	15.1	12.6	9.7	6.0	2.2
	5%	DB	3.5	4.0	6.8	11.2	14.9	18.2	20.3	20.4	17.1	13.3	9.5	5.0
		MCWB	1.0	1.4	3.3	6.8	11.0	14.2	14.2	14.1	11.7	9.0	5.4	2.1
	10%	DB	2.0	2.2	5.1	9.2	13.4	16.8	18.7	19.0	15.4	11.7	7.9	3.3
		MCWB	0.3	0.6	2.5	5.9	10.1	13.5	14.0	13.7	11.0	8.1	4.6	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	4.7	6.9	10.2	14.3	17.5	17.5	17.6	14.6	13.0	9.3	5.6
		MCDB	5.9	8.2	10.3	13.5	16.8	19.6	20.0	20.5	18.9	15.8	12.8	6.2
	2%	WB	2.4	2.9	5.1	8.7	12.6	16.1	16.2	16.2	13.4	11.0	7.7	4.1
		MCDB	3.8	5.2	7.9	12.1	15.3	18.7	19.5	19.5	17.3	13.4	9.8	5.3
	5%	WB	1.4	1.8	3.9	7.5	11.6	14.8	15.4	15.4	12.5	9.9	6.6	2.7
		MCDB	2.7	3.1	6.0	10.4	14.1	17.4	18.5	18.7	15.9	12.1	8.4	4.1
	10%	WB	0.4	0.8	2.8	6.3	10.6	13.8	14.6	14.5	11.6	8.9	5.5	1.6
		MCDB	1.7	2.0	4.9	9.0	12.8	16.4	17.5	17.8	14.5	10.9	6.9	2.9
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.6	4.9	5.2	5.2	5.2	5.6	5.4	5.1	5.1	4.4	4.3
		MCDBR	5.5	6.1	6.6	7.5	7.2	6.7	7.8	6.9	7.3	6.9	5.8	5.1
		MCWBR	4.1	4.1	4.2	4.7	5.1	5.1	4.8	4.3	4.0	4.3	3.7	3.7
	5% WB	MCDBR	4.7	5.0	6.0	6.6	6.6	6.0	6.3	6.0	6.3	6.3	5.0	4.3
		MCWBR	3.9	3.9	4.3	4.7	5.3	5.2	4.9	4.5	4.2	4.4	3.7	3.7
Clear-Sky Solar Irradiance	taub	0.260	0.280	0.312	0.352	0.345	0.345	0.347	0.354	0.335	0.306	0.280	0.262	
	taud	2.468	2.445	2.428	2.324	2.378	2.393	2.378	2.360	2.413	2.500	2.546	2.507	
	Ebn at Noon	899	932	935	914	926	924	917	901	896	887	868	865	
	Edh at Noon	71	87	101	121	118	117	118	116	102	82	67	63	
All-Sky Solar Radiation	RadAvg	1.58	2.39	3.43	4.51	5.41	6.09	6.29	5.68	4.17	2.83	1.77	1.32	
	RadStd	0.24	0.31	0.36	0.53	0.45	0.50	0.47	0.48	0.33	0.35	0.20	0.19	